12TH CHEMISTRY · CHAPTER 18 · CHAPTER TEST

Transition Metals

Chapter-wise test: 6 MCQs, 5 of 6 short questions, 1 long question

Class: 12th (HSSC Part II)	**Subject:** Chemistry	**Total Marks:** 26
Chapter: 18	**Time Allowed:** 45 minutes	**Name:** _____

OBJECTIVE PART (6 × 1 = 6)

Q.1 Choose the correct answer.

1. Which 3d element shows the highest oxidation state (+7)?

(a) Cr (b) Fe
(c) Cu (d) Mn

2. The coordination number in an octahedral complex is:

(a) 2 (b) 6
(c) 4 (d) 8

3. The geometry with a bond angle of 109.5° is:

(a) Tetrahedral (b) Square planar
(c) Octahedral (d) Linear

4. The colour of transition metal complexes is due to:

(a) s-p transition (b) p-p transition
(c) d-d transition (d) s-s transition

5. Cobalt(II) ions with conc. HCl give a solution that is:

(a) Pink (b) Blue
(c) Yellow (d) Green

6. Which ligand is hexadentate?

(a) Ammonia (b) Water
(c) EDTA (d) Cyanide ion

SUBJECTIVE PART — SHORT QUESTIONS (5 × 2 = 10)

Q.2 Write short answers to any FIVE (5) questions.

i. Why is zinc not classified as a transition element?
ii. Why do chromium and copper show anomalous electronic configurations?
iii. Define a ligand and coordination number.
iv. What is a d-d transition?
v. Define the stability constant of a complex.
vi. Why is cisplatin polar but transplatin non-polar?

LONG QUESTION (2 × 5 = 10)

Q.3 Answer both parts.

(a) Describe d-orbital splitting in octahedral and tetrahedral complexes.
(b) Explain cis-trans and optical isomerism in complexes.

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